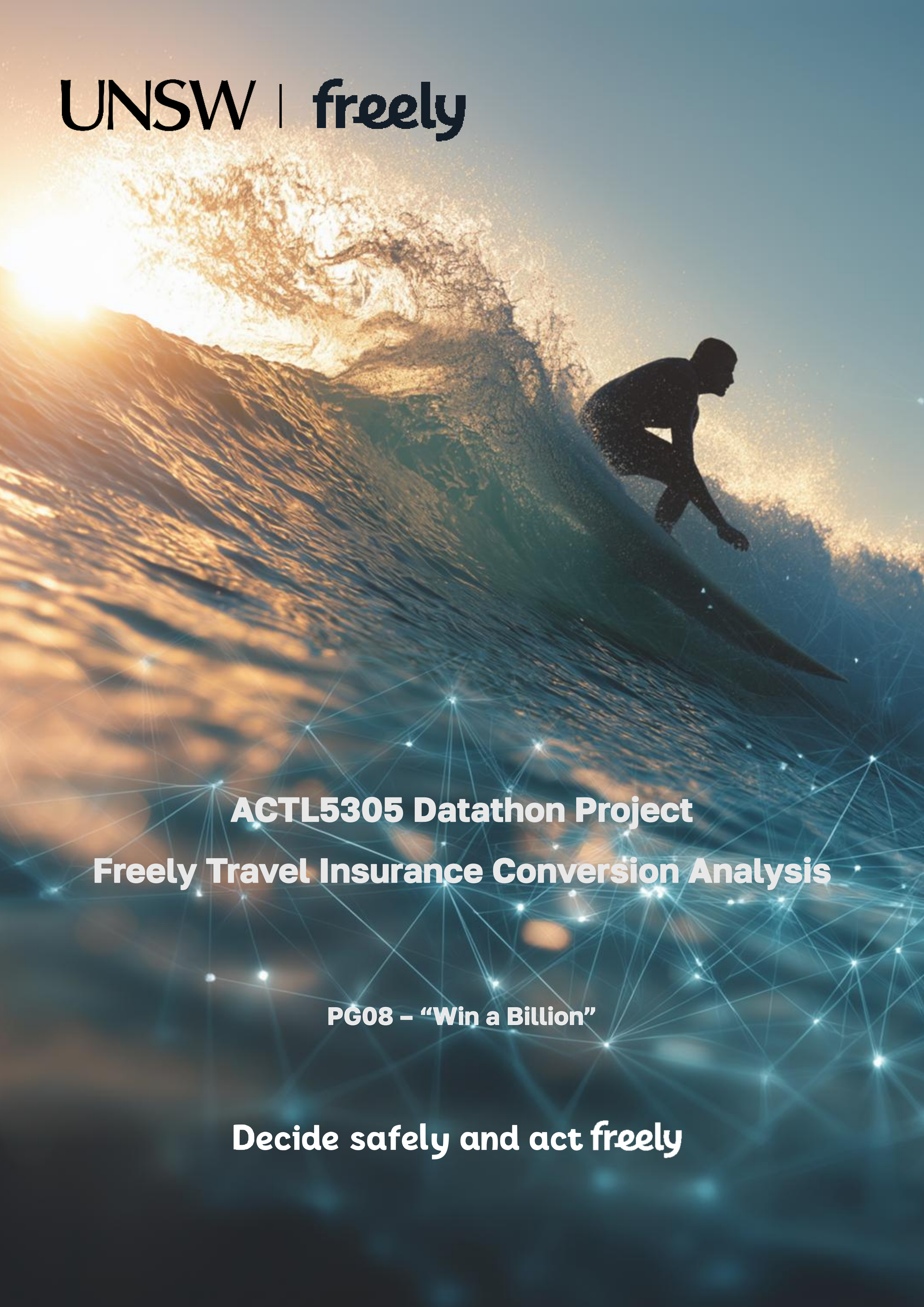

UNSW | freely

ACTL5305 Datathon Project

Freely Travel Insurance Conversion Analysis

PG08 – “Win a Billion”

Decide safely and act freely

Executive Summary

This project analyzes Freely's quote-to-conversion data to identify the key predictors of purchase intent, quantify behavioural patterns in optional boost selections, and provide actionable recommendations to improve conversion and revenue efficiency. Our analysis integrates feature engineering, predictive modelling, evaluation, and business interpretation to develop a transparent and replicable analytical framework.

After correcting ambiguous date fields using a hybrid machine learning method, 71 variables (including 9 innovated variables) were developed to capture demographic, temporal, product, destination, and behavioural dimensions. We compared several approaches in predictive modelling, including Generalized Linear Model (fitted under no regularization, Lasso, Ridge and Elastic Net), LightGBM, XGBoost, neural networks with embedded layer and Tabnet. Several balancing method, like over- and under-sampling, class weighting and focal loss were utilized to mitigate imbalance problems. Then, model performance was evaluated using the Precision–Recall Area Under Curve (PR-AUC), selected for its suitability in imbalanced conversion data. XGBoost achieved the best PR-AUC, identifying around 77.6% of converted quotes in the test set, while the GLM was retained for interpretability and business interpretation.

Using GLM, AMEs were estimated to measure each variable's average effect on conversion probability (Δp, in percentage points), and an AME–Coverage matrix was constructed to segment customers by effect magnitude and population share. Our analysis identifies clear behavioral and contextual drivers of conversion. Younger and solo travelers show the highest likelihood of purchase, while families with children tend to have lower completion due to coordination complexity and cost sensitivity. Customers who use the app, quote and purchase on the same day, or specify realistic and focused destinations rather than listing many scattered locations are much more likely to convert, reflecting genuine purchase intent. Pricing, perceived risk, and market context also influence behaviour. Discounts strongly raise conversion, whereas higher daily prices and negative government travel advisories reduce it. When exchange rate volatility increases, travelers tend to confirm purchases earlier to secure cost certainty. Boost selections further signal readiness to buy. Existing Medical, Gadget, and Snow Sports boosts raise conversion, while Extra Cancellation indicates hesitation. Selecting two or more boosts consistently increases conversion, showing that proactive coverage customization reflects stronger commitment. Overall, immediacy, value perception, and proactive protection emerge as the key determinants of customer conversion decisions.

The GLM–AME framework further enables ongoing monitoring. The GLM can be retrained every one to two years to capture structural behavioural changes, while AMEs can be refreshed monthly to track short-term ROI and the effectiveness of marketing or pricing strategies without rebuilding the full model.

Based on the AME-Coverage analysis, three strategic actions are recommended. First, direct investment toward high-uplift and low-coverage segments such as mid-aged solo travelers and activity-oriented boosts, complemented by seasonal and bundled offers such as Snow-plus-Gadget or Medical-plus-Cruise. Second, improve efficiency in high-coverage but low-uplift areas such as family segments and domestic destinations by simplifying the purchase flow, clarifying multi-child coverage, and refining message targeting rather than expanding marketing spend. Third, use behavioral and market triggers such as exchange-rate spikes, government advisories, evening and weekend activity peaks, and same-day quotes to increase conversion. At the micro level, Freely could run randomized A/B tests, event studies, and difference-in-differences analyses to confirm causal impact and guide continuous optimization.

1 Introduction

In the travel insurance market, conversion rate (the proportion of quotes that turn into completed purchases) is a critical performance metric linking marketing effectiveness to revenue generation. Even a marginal uplift in conversion can translate into substantial financial gains without additional advertising expenditure. We were engaged by Freely, an emerging travel insurance brand backed by Zurich Insurance Group, to conduct a comprehensive predictive and impact-factor analysis aimed at improving its quote-to-conversion performance.

This report follows a structured data-science workflow. Section 2 details how we transformed the raw Freely dataset into an analytically reliable form. Section 3 presents the predictive models, compares alternative algorithms, and evaluates their performance using appropriate imbalance-aware metrics. Section 4 interprets the model results to identify drivers of conversion and provide boost insights. Finally, Section 5 translates these findings into business recommendations, and outlines additional data and experimental methods required to establish causal inference.

2 Data Cleaning and Feature Engineering

Freely's dataset's main problem is the inconsistent date formats. Data cleaning involved two steps: logical consistency checks and ML-based inference. We employed three rules to identify valid combinations: (i) Quote date < departure date. (ii) Departure date < return date. (iii) Boost coverage dates need to be within the trip duration (correct date format for 66.9% of the records, 34 records were omitted). For the remaining ambiguous cases, a random forest model was trained on the verified subset to learn the relationship between trip duration and behavioural features such as discounts, group size, and booking patterns (effectively resolving 33.1% of the previously uncertain cases). We engineered **62 basic variables**, as summarized in Appendix A with five main categories consisting of **demography (18 variables)**, **time (5 variables)**, **product (6 variables)**, **popular destination and continent (18 variables)**, **boosts (15 variables)** as well as interaction terms. Then, we developed **9 innovative variables** (see Appendix A), with detailed calculation procedures provided in Appendix B. These variables are designed to capture critical dimensions of travel behavior, such as **temporal aspects**, **authenticity of experience**, and **destination-specific risk factors**.

3 Modelling

For the impact factor analysis, we adopted only the Generalized Linear Model (GLM) due to its interpretability and ability to reveal the direction of each effect. For the predictive modelling task, we compared methods including Lasso GLM, Ridge GLM, Elastic Net GLM, LightGBM, XGBoost, a feed-forward neural network with embedding layers (see Appendix C for the architecture), and TabNet (Arik and Pfister 2020).

3.1 Multicollinearity, Data Splitting and Handling Class Imbalance

Multicollinearity was addressed only for GLMs. During feature engineering, redundant variables were avoided. Then, we removed highly correlated variables. For instance, is_bali was found to be highly correlated with the detailed destination indicator and was excluded. Lastly, if the estimated coefficient signs were inconsistent with domain knowledge, the corresponding variables were reviewed. After adjusting, the final pairwise correlations are less than 0.35.

For predictive modelling, the dataset was split 80/20 with 20% of training data for validation. Given the low conversion rate (12.85%), we implemented targeted balancing strategies to prevent model bias. **Over- and under-sampling** were used to equalize the training data for neural networks, **class weighting** was applied in GLM and tree-based models to highlight converted customers, and **focal loss** (Lin et al. 2017) was introduced to neural models to strengthen learning from hard-to-detect cases. These measures improved converted customer identification across models.

3.2 Model Comparison

In Freely's case, traditional ROC-AUC may appear artificially high under class imbalance. Therefore, we adopted the more robust **Precision–Recall Area Under Curve (PR-AUC)** as the principal evaluation metric. Gradient boosting methods (XGBoost and LightGBM) achieved the highest PR-AUC scores (0.4575 and 0.4571, respectively), outperforming both linear and neural models while maintaining moderate interpretability. Neural networks with over- and under-sampling obtained slightly lower PR-AUC values (0.4288 and 0.4183), suggesting that rebalancing strategies and focal loss effectively improved detection of rare conversions. Among linear baselines, the Elastic Net performed best (0.3957), followed closely by Lasso and Ridge, see Appendix Figure 2 for more details.

In conversion modelling, identifying converted cases is of business importance. Hence, **recall** was prioritized over overall accuracy, as it directly reflects the model's ability to detect positive (converted) samples. Specifically, our models successfully identified nearly 80% of converted samples in the test set, indicating strong recall. However, some non-converted samples were incorrectly classified as converted, leading to a low precision (as illustrated in Appendix Figure 3). In practice, each predicted conversion represents potential revenue, but excessive false positives could increase marketing costs, calling for a balanced trade-off between recall and precision. The boosting models (XGBoost and LightGBM) provided the most balanced and robust outcomes, achieving higher precision without a significant loss in recall. They also offer moderate interpretability, which facilitates communication with business teams. **Accordingly, we selected XGBoost for recommendation and dashboard visualization.** If the business objective shifts toward maximizing recall with minimal concern for precision (e.g., in low-cost marketing campaigns), feedforward neural networks and TabNet with stronger positive-class weighting could be further explored.

4 Key Predictors and Boost Insights

The relative importance of predictors was assessed using GLM-based marginal probability changes (Δp, pp, see Appendix F), following the procedure in Appendix E. In parallel, exploratory data analysis (EDA) was conducted to identify customer segments more inclined to select additional coverages ("boosts").

4.1 Customer and Group Composition

Customer demographics and group structure exert strong influences on conversion probability. A higher **child ratio** shows a positive marginal effect (+4.63 pp), suggesting that families with more children are more motivated to secure travel insurance. Conversely, the **senior ratio** contributes a small negative effect (–0.30 pp). This aligns with previous research (Wang et al. 2021) showing that families in developed countries prioritize insurance purchases for their children's safety but elderly travelers face higher base premiums and medical disclosure complexity, discouraging completion. Traveler type is another dominant factor: **Solo travelers** exhibit the strongest positive effects, reflecting simpler decision-making and clearer intent. By contrast, **family travelers with children** show negative coefficients, implying lower completion likelihood due to coordination complexity and higher cost sensitivity. Overall, conversion is higher for younger and more independent customer segments, which aligns with Freely's positioning as a **youth-oriented brand** appealing to spontaneous and experience-driven travelers.

4.2 Genuity Matters: Signals of Real Purchase Intent

Not every quote reflects genuine purchase intention. Our analysis shows that customers demonstrating stronger commitment are much more likely to complete their purchase. The **app** platform increases conversion by +19.16 pp relative to the website baseline, suggesting that users who go beyond browsing and download the app show greater intent to buy. Similarly, **same-day quotes** increase conversion by +10.41 pp, reflecting time urgency and concrete travel plans. Travelers specifying **precise destinations** (+2.6 pp) also exhibit higher planning confidence, confirming that those with well-defined itineraries are more committed. In contrast, customers with **unrealistically dispersed destinations** are less likely to convert, as such quotes often indicate casual or exploratory browsing. Taken together, these results show that genuine and purposeful engagement remains the strongest signal of conversion.

4.3 Situation Perception and Human Behaviour

Conversion behaviour also reflects travelers' perceptions of risk, cost, and situational confidence. A higher **price per person per day** slightly reduces conversion (–0.29 pp), while **discounts** have a strong, monotonic positive impact (+11.4 to +27.9 pp from 10% to 20%), indicating that travelers respond sensitively to perceived value. Temporal variables provide further evidence: quotes made during **evening hours** and **weekends** are slightly more likely to convert, consistent with leisure-time decision windows, while quotes issued in **late November to early December** show a moderate decline (–2.9 pp), likely reflecting pre-holiday budget constraints. Government advisories significantly deter purchases, with "**do not travel**" and "**reconsider travel**" categories lowering conversion by around –3.5 pp. Interestingly, macro-financial factors act in the opposite direction: when **extreme exchange rate risk** is high, travelers tend to confirm purchases earlier (+0.9 pp), likely seeking cost certainty. Regional preferences also reflect perceived affordability, with destinations such as **Europe during the AUS autumn/winter** (–3.4 pp), as well as **Australia** and **USA** (–3.4 and –2.6 pp) showing lower conversion. These findings illustrate that conversion is shaped not only by product design but also by how travelers perceive economic and safety risks in their environment.

4.4 Boosts as Behavioural Signals of Proactive Protection

Customers who actively customize coverage are generally more serious about their trips and aware of potential risks, turning boost selection into a strong indicator of purchase readiness.

Who purchases boosts? Exploratory analysis (see Appendix G) shows clear profile patterns: younger travelers (19–34) favor activity-based or technology-related boosts such as Snow Sports, Adventure Activities, Gadget, and Motorcycle, while older customers (59–79) prefer Existing Medical and Cruise Cover. Smaller groups and shorter trips tend to add more boosts, and destinations like Japan or Canada are closely associated with snow-related covers, whereas Bali and Indonesia align with motorcycle and gadget protection. Seasonal variation reinforces these trends. Australia's summer and autumn coincide with the northern winter, driving higher demand for snow-related boosts, while spring corresponds to increased motorcycle cover purchases.

How do boosts affect conversion? Boost choices correlate strongly with conversion outcomes. Selecting at least two boosts (**num_boosts2+**) raises conversion by +4.35 pp, while specific types, particularly **Existing Medical** (+19.99 pp), **Gadget** (+8.49 pp), and **Snow Sports** (+7.86 pp), produce the highest uplifts. In contrast, uncertainty-related boosts such as **Extra Cancellation** (–7.65 pp) and **Cruise** (–1.96 pp) indicate hesitation. These findings confirm that proactive, activity-based boosts enhance conversion, whereas optional cancellation coverage signals lower commitment.

5 Business Analysis

This section identifies key business growth opportunities through Average Marginal Effect, customer percentage (**coverage**), and Return on Investment (ROI) analysis (see Appendix H). By combining data-driven insights with marketing strategies, we outline macro-level investment priorities and micro-level actions to enhance overall conversion performance. Furthermore, potential causal effects are briefly explored.

5.1 Macro-Level Strategy: Growth Priorities

Our AME–Coverage matrix highlights several promising segments (see Appendix H.3). Segments with high AME but low coverage represent underpenetrated yet highly responsive customers — ideal targets for expansion. Investing in these 'high-uplift, low-reach' groups can yield stronger ROI, consistent with marginal utility theory in marketing investment (Farris et al. 2010) and (Rust et al. 2004). For example, promoting solo travelers aged 40–59, increasing motorcycle, adventure, and gadget cover boosts, and cruise marketing in Australia are likely to deliver outsized conversion gains (see Appendix H.4.3). Freely already performs well in High Uplift & High Coverage (see Appendix H.4.3), indicating that current marketing strategies in these areas are both effective and sustainable.In contrast, segments with negative AME but high coverage, such as family trips with children or popular destinations like Australia and the US, show low marketing efficiency. For these, further investment would reduce ROI; instead, process optimization (e.g., simplifying purchase flows) or message refinement (e.g., improving content and targeting) should be prioritized.

5.2 Micro-Level Tactics: Suggestions and Casual Effects

5.2.1 Product and Boosts

- For low-conversion boosts such as Extra Cancellation (AME –13.2 pp), **defer their display** until later in the quote flow to avoid early hesitation. Future validation could randomize the quote-flow position of *Extra Cancellation* and apply difference-in-differences (DID) to test whether early exposure causally reduces conversion.

- **Apply bundle pricing** for high-intent people (e.g., Medical + Cruise; Snow + Gadget) to lift the revenue.
- **Introduce context-aware boost suggestions** based on destination, age, and season (e.g., Japan ski trips → Snow Sports + Gadget). This can reduce search friction and align with travellers' planning intent.
- **Use positive language** to strengthen incentives for high-impact boosts such as **Existing Medical** (AME +18.4 pp), **Gadget Cover** (AME +8.6 pp), and **Snow Sports** (AME +7.9 pp).
- **Run targeted seasonal campaigns** (e.g., "Heading to Japan this December? Don't forget Snow Sports cover.") to match coverage relevance with travel timing.

5.2.2 Customer and Timing

- **Prioritize solo travelers** (AME +12 to +15 pp) as the most ROI-efficient audience for early-stage marketing. For family segments (especially those with children), **reduce friction and clarify multi-child coverage**, as those families show higher motivation when benefits are explicit (*child_ratio* AME +4.63 pp).
- **Push completion in the app** (AME +17.57 pp). To test whether an app migration prompt truly increases conversion, Freely could run a randomized A/B experiment with intent-balanced groups. Record assignment, exposure, click, and completion timestamps, and compare pre-prompt behaviour to detect intent bias. Include a small holdout group among high-discount users to isolate the pure prompt effect.
- **Emphasize quick checkout** for same-day quotes (AME +10.41 pp).
- **Time-sensitive opportunities:** Leverage real-time offers for same-day quotes (AME +10.32 pp) and increase bids during evening hours (19:00–22:00, AME +0.56 pp) or weekends (AME +1.00 pp) when activity peaks. During low-conversion periods such as late November–early December (AME –3.70 pp), replace large monetary discounts with early-bird or "price lock" messaging to preserve value.
- **Responsive marketing triggers:** Integrate dynamic pricing or message shifts during macro events. When exchange rate volatility spikes (*extreme_exchange_risk_30d*, AME +0.91 pp), promote "lock your travel budget" messages; if negative government advisories are issued (AME≈-4.3 pp), shift communication towards safety and medical upgrades instead of direct selling. To test the causality, Freely is suggested to use an event-study or DID framework, with unaffected destinations as controls. This analysis requires daily logs of destination, quote time, and user segment to identify behavioral shifts.

Thank you for reading!

We appreciate your time and interest. You can explore our interactive dashboard on the R Shiny server.

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A Variables List

Table 1: Variables List

Number	Variable	Type	Variable Name in Coding
1	Couple (18-29)	Demographic	age_and_relationshipCouple (18-29)
2	Couple (30-39)	Demographic	age_and_relationshipCouple (30-39)
3	Couple (40-49)	Demographic	age_and_relationshipCouple (40-49)
4	Couple (60-69)	Demographic	age_and_relationshipCouple (60-69)
5	Couple (70+)	Demographic	age_and_relationshipCouple (70+)
6	Family (Children 10-18)	Demographic	age_and_relationshipFamily (Children 10-18)
7	Family (Children <10)	Demographic	age_and_relationshipFamily (Children <10)
8	Family (Children >18)	Demographic	age_and_relationshipFamily (Children >18)
9	Other	Demographic	age_and_relationshipOther
10	Solo (18-29)	Demographic	age_and_relationshipSolo (18-29)
11	Solo (30-39)	Demographic	age_and_relationshipSolo (30-39)
12	Solo (40-49)	Demographic	age_and_relationshipSolo (40-49)
13	Solo (50-59)	Demographic	age_and_relationshipSolo (50-59)
14	Solo (60-69)	Demographic	age_and_relationshipSolo (60-69)
15	Solo (70+)	Demographic	age_and_relationshipSolo (70+)
16	Group Size	Demographic	group_size
17	Senior Ratio	Demographic	senior_ratio
18	Child Ratio	Demographic	child_ratio
19	Trip Duration	Time	num_days
20	Quote Hour (in 19:00-22:00)	Time	quote_hour_19_22
21	Quote Date (in 26/11 - 09/12)	Time	quote_late_nov_early_dec
22	Is weekend	Time	is_weekend
23	Are holidays	Time	is_holidays
24	APP	Product	platformapp
25	QW	Product	platformqw
26	Price per traveler per day	Product	price_per_person_per_day
27	Discount 10%	Product	discount_10
28	Discount 15%	Product	discount_15
29	Discount 20%	Product	discount_20
30	Japan	Popular Destination	has_japan
31	Malaysia	Popular Destination	has_malaysia
32	Spain	Popular Destination	has_spain
33	USA	Popular Destination	has_usa
34	Italy	Popular Destination	has_italy
35	New Zealand	Popular Destination	has_nz
36	India	Popular Destination	has_india
37	Indonesia	Popular Destination	has_indonesia
38	France	Popular Destination	has_france
39	Australia	Popular Destination	has_australia

Table 1 (continued)

Number	Variable	Type	Variable Name in Coding
40	Europe in Aus Autumn & Winter	Popular Destination - Seasonality	europe_in_aus_autumn_winter
41	UK in Aus Winter	Popular Destination - Seasonality	uk_in_aus_winter
42	Italy in Aus Winter	Popular Destination - Seasonality	italy_in_aus_winter
43	Canada in Aus Winter	Popular Destination - Seasonality	canada_in_aus_winter
44	Spain in Aus Winter	Popular Destination - Seasonality	spain_in_aus_winter
45	Antarctica	Continent	is_antarctica
46	Africa	Continent	is_africa
47	South America	Continent	is_south_america
48	Number of boosts: 1	Boosts - Indicators	num_boosts1
49	Number of boosts: 2+	Boosts - Indicators	num_boosts2+
50	Extra cancellation	Boosts - Indicators	extra_cancellation_boost
51	Specified item	Boosts - Indicators	specified_item_boost
52	Gadget cover	Boosts - Indicators	gadget_cover_boost
53	Existing medical	Boosts - Indicators	existing_medical_boost
54	Adventure activities	Boosts - Indicators	adventure_activities_boost
55	Snow sport	Boosts - Indicators	snow_sport_boost
56	Rental vehicle	Boosts - Indicators	rental_vehicle_boost
57	Cruise	Boosts - Indicators	cruise_boost
58	Motorcycle	Boosts - Indicators	motorcycle_boost
59	Extra cancellation with senior	Boosts - Interaction	extra_cancellation_with_senior
60	Snow sport with children	Boosts - Interaction	snow_sport_with_children
61	Snowsport in Canada	Boosts - Interaction	snow_sport_in_canada
62	Cruise in Australia	Boosts - Interaction	cruise_in_australia
63	Destination: Reconsider your need to travel	Innovated	warning_reconsider_travel
64	Destination: Do not travel	Innovated	warning_do_not_travel
65	Booking Advance Ratio	Innovated	booking_advance_ratio
66	Same Day of Quote and Start	Innovated	same_day
67	Very Precise description of destinations	Innovated	is_very_precise_destination
68	Maximum pairwise distances in multi-destination (short distance)	Innovated	tds_distance_short
69	Maximum pairwise distances in multi-destination (medium distance)	Innovated	tds_distance_medium
70	Maximum pairwise distances in multi-destination (long distance)	Innovated	tds_distance_long
71	Extreme Exchange Rate Risk	Innovated	extreme_exchange_risk_30d

B Calculation Method of Some Innovated Variables

This appendix provides the computational details of the innovative variables described in Appendix A. All variables are designed to capture behavioral and contextual signals indicating the genuineness and conversion likelihood of each quote request.

B.1 Booking Advance Ratio

Let d_q denote the quote creation date, and d_s the trip start date. Let T represent the trip duration (in days). The booking advance ratio is defined as:

$$\text{Booking Advance Ratio} = \frac{(d_s - d_q)}{T}. \tag{1}$$

A higher ratio suggests that the customer plans the trip well in advance, while a lower ratio indicates a short notice or impulsive purchase.

B.2 Very Precise Description of Destinations

A destination string is classified as *very precise* if it contains geographic markers below the country level, such as state, city, specific attractions, or subregions.

B.3 Maximum Pairwise Distance in Multi-Destination Trips

For quotes listing multiple destinations, we compute the maximum pairwise geodesic distance between any two coordinates. Let $\{(lat_i, lon_i)\}_{i=1}^{n}$ denote the coordinates of n destinations in a quote. The maximum pairwise distance is given by:

$$D_{\max} = \begin{cases} 0, & n = 1, \\ \max\limits_{1 \le i < j \le n} \text{GeoDist}\big((lat_i, lon_i), (lat_j, lon_j)\big), & n > 1, \end{cases} \tag{2}$$

where GeoDist$(\cdot)$ is the great-circle distance (Haversine formula). Customers specifying unrealistically long distances among destinations may be less likely to finalize a purchase.

To capture different levels of travel dispersion, we further classify each quote into one of three categories according to $D_{\max}$:

$$\text{tds\_distance\_short} = \begin{cases} 1, & 1 < D_{\max} \le 1000, \\ 0, & \text{otherwise}; \end{cases} \tag{3}$$

$$\text{tds\_distance\_medium} = \begin{cases} 1, & 1000 < D_{\max} \le 5000, \\ 0, & \text{otherwise}; \end{cases} \tag{4}$$

$$\text{tds\_distance\_long} = \begin{cases} 1, & D_{\max} > 5000, \\ 0, & \text{otherwise}. \end{cases} \tag{5}$$

Here, distances are measured in kilometers. The threshold values of 1000 km and 5000 km approximately separate short-haul (intra-regional), medium-haul (intercontinental within one hemisphere), and long-haul (cross-hemisphere) trips. Notably, the definition of tds_distance_short starts from 1 km rather than 0 to prevent perfect collinearity.

B.4 Destination Warning

We matched all destinations against the Australian Government's official travel advisory portal. Each destination is assigned one of the four advisory levels: *Exercise normal safety precautions*, *Exercise a high degree of caution*, *Reconsider your need to travel*, and *Do not travel*. We define two binary indicators:

$$\text{warning\_reconsider\_travel} = \begin{cases} 1, & \text{if advisory = "Reconsider your need to travel"}, \\ 0, & \text{otherwise;} \end{cases} \tag{6}$$

$$\text{warning\_do\_not\_travel} = \begin{cases} 1, & \text{if advisory = "Do not travel"}, \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

B.5 Extreme Exchange Rate Risk

We use daily exchange rate data from the exchangerate-api to quantify currency volatility faced by customers. These exchange rates are relative to the Australian Dollar (AUD).

For each quote date d_q, and for each currency $c \in C_{\text{quote}}$ associated with the customer's destinations, let $r_t^{(c)}$ be the exchange rate on day t. Define the daily percentage change as:

$$\Delta_t^{(c)} = \frac{r_t^{(c)} - r_{t-1}^{(c)}}{r_{t-1}^{(c)}} \times 100, \tag{8}$$

the 30-day exchange rate volatility prior to d_q is then:

$$\sigma_{30}^{(c)} = \sqrt{\frac{1}{29} \sum_{t=d_q-29}^{d_q} \left(\Delta_t^{(c)} - \bar{\Delta}^{(c)}\right)^2}. \tag{9}$$

For quotes involving multiple currencies, we take the maximum volatility across all destinations:

$$\text{exchange\_volatility\_30d} = \max_{c \in C_{\text{quote}}} \sigma_{30}^{(c)}. \tag{10}$$

Finally, the variable extreme_exchange_risk_30d is defined as:

$$\text{extreme\_exchange\_risk\_30d} = \begin{cases} 1, & \text{if exchange\_volatility\_30d} > P_{90}, \\ 0, & \text{otherwise,} \end{cases} \tag{11}$$

where P_{90} denotes the 90th percentile of all observed 30-day volatilities across quotes.

C Feed-Forward Neural Network Structure

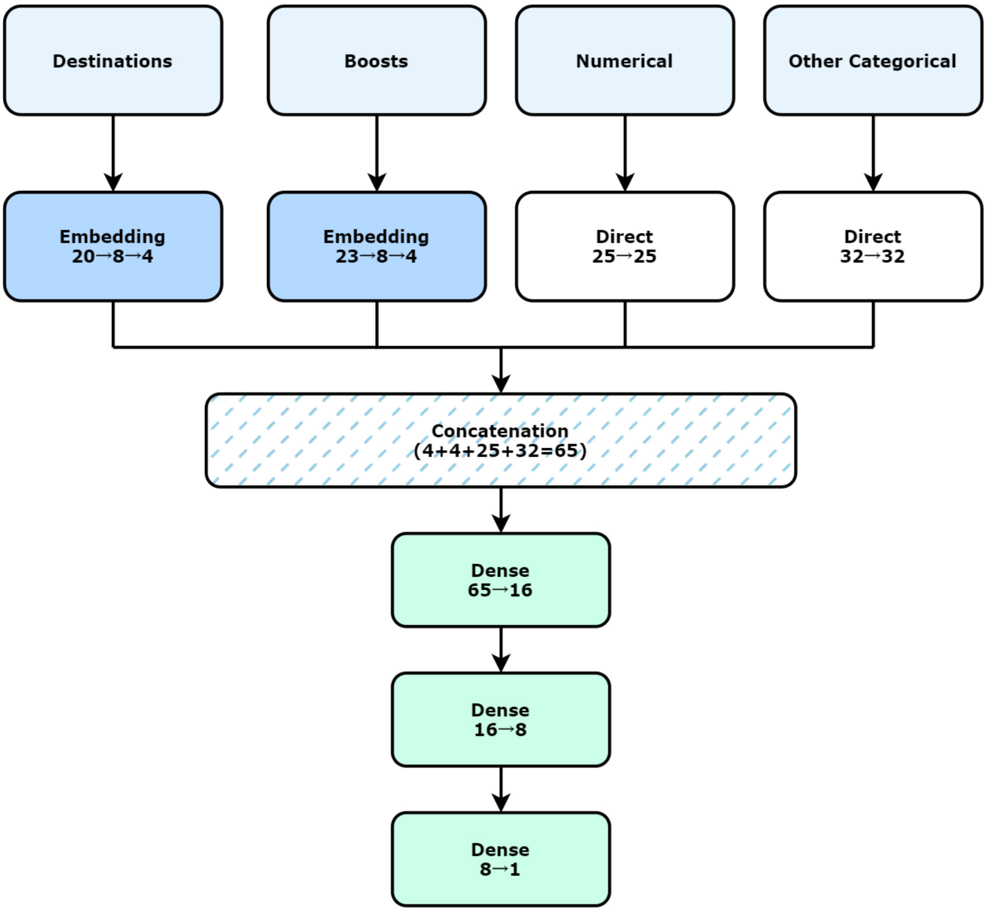

Figure 1: Feed-forward neural network structure. Batch normalization and dropout layers are included to reduce overfitting.

D Figure of Model Comparison and Precision-Recall Trade Off

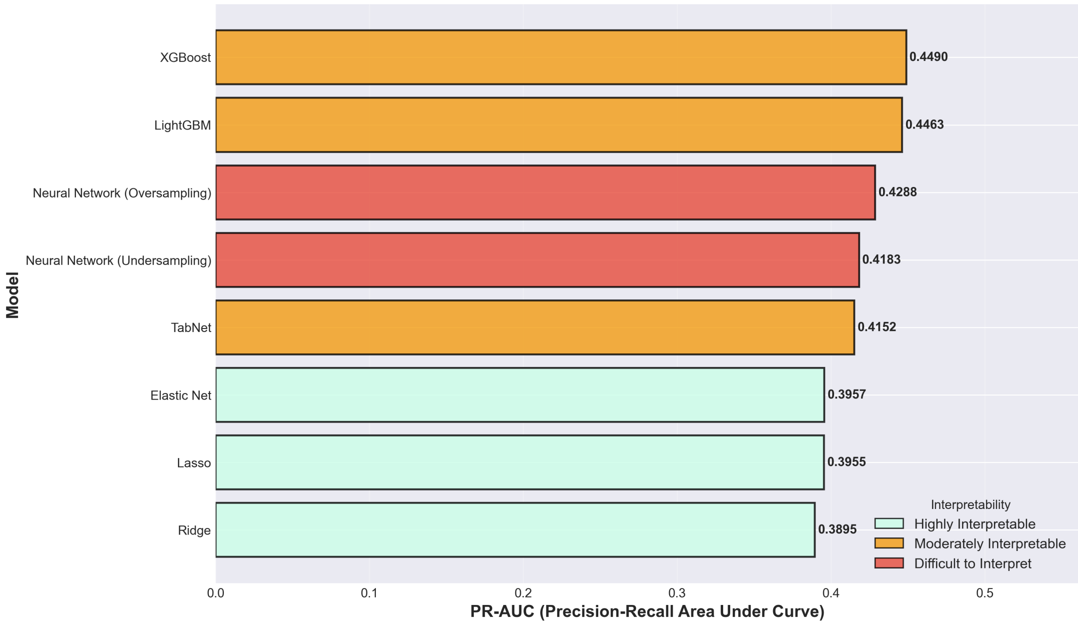

Figure 2: Model comparison based on PR-AUC.

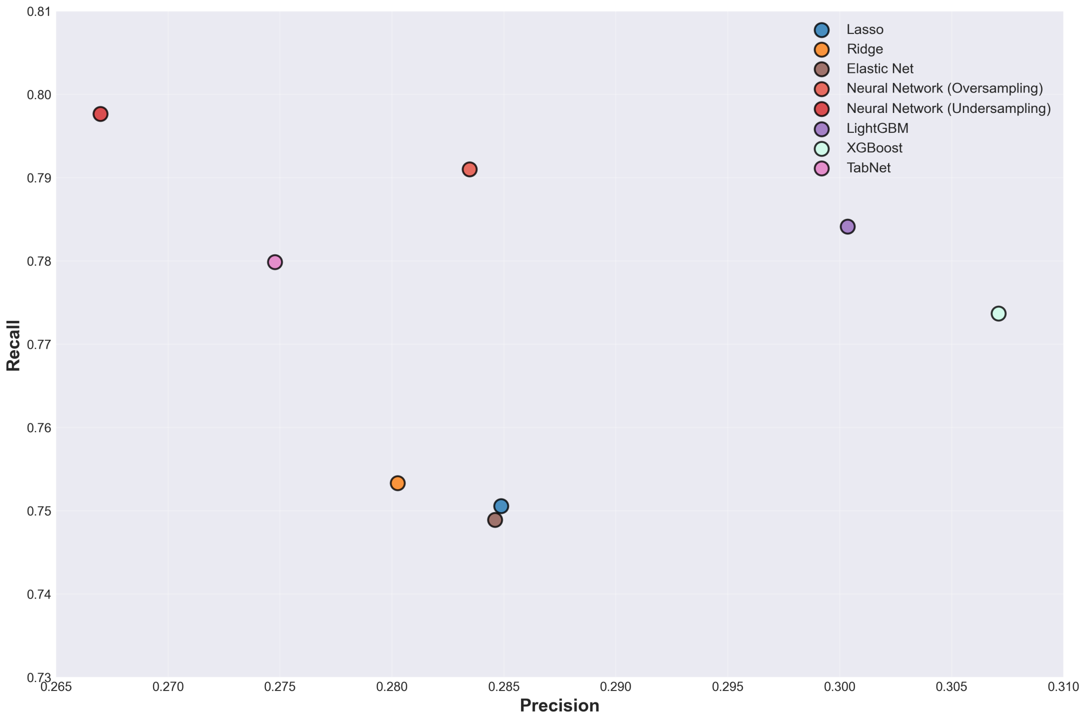

Figure 3: Precision–Recall trade-off across models.

E Methodology for GLM Interpretation

E.1 Model Specification

The logistic Generalised Linear Model (GLM) was fitted to estimate the probability of conversion. For each quote i,

$$\eta_i = \beta_0 + \sum_j \beta_j x_{ij}, \qquad p_i = \frac{1}{1 + e^{-\eta_i}}.$$

The estimated intercept was $\beta_0 = -2.32845$. Predictors include demographic, trip, product, destination, and boost variables (see Appendix A) for definitions.

E.2 Marginal Probability Change (Δp)

For each predictor x_j, the marginal probability effect Δp_j quantifies how conversion probability changes when x_j shifts by one unit (for continuous variables) or from 0 to 1 (for binary variables), holding other variables constant:

$$\Delta p_j = p(X, x_j = 1) - p(X, x_j = 0).$$

For small continuous changes Δx, a first-order approximation is used:

$$\Delta p_j \approx p(1 - p)\beta_j \Delta x.$$

All reported values are expressed in percentage points (pp) and calculated at the model baseline $\eta = \beta_0$.

E.3 Interpretation

A positive Δp_j implies that the predictor increases the likelihood of purchase, while a negative value indicates a reduction. These effects are conditional on the model structure and differ from raw descriptive conversion rates. For example, a Δp_j of +10 pp means that switching the variable from 0 to 1 increases the predicted probability of purchase by ten percentage points, assuming all other factors remain constant.

F GLM-Based Marginal Probability Effects

Table 2: Model Coefficients and Marginal Effects Summary

Variable	Beta	ΔX	Δp	Δp (pp)
discount_20	1.7851	1	0.2786	27.862
existing_medical_boost	1.4268	1	0.1999	19.992
platformapp	1.3861	1	0.1916	19.164
age_and_relationshipSolo (70+)	1.2644	1	0.1677	16.775
discount_15	1.2441	1	0.1639	16.390
age_and_relationshipSolo (40–49)	1.1661	1	0.1495	14.945
age_and_relationshipSolo (60–69)	1.0955	1	0.1369	13.688
age_and_relationshipSolo (50–59)	1.0746	1	0.1332	13.324
age_and_relationshipSolo (18–29)	1.0685	1	0.1322	13.219
age_and_relationshipSolo (30–39)	1.0065	1	0.1217	12.170
discount_10	0.9595	1	0.1140	11.400
Same_day	0.8972	1	0.1041	10.411
cruise_in_australia	0.7922	1	0.0883	8.829
gadget_cover_boost	0.7686	1	0.0849	8.488
snow_sport_boost	0.7240	1	0.0786	7.857
extra_cancellation_boost	-2.0590	1	-0.0765	-7.652
motorcycle_boost	0.7025	1	0.0756	7.559
extra_cancellation_with_senior	0.4821	1	0.0475	4.751
adventure_activities_boost	0.4685	1	0.0459	4.592
age_and_relationshipFamily (Children <10)	-0.7495	1	-0.0448	-4.477
num_boosts2+	0.4475	1	0.0435	4.349
child_ratio	2.6575	0.1667	0.0430	4.296
age_and_relationshipFamily (Children 10–18)	-0.6157	1	-0.0388	-3.878
warning_do_not_travel	-0.5486	1	-0.0355	-3.550
warning_reconsider_travel	-0.5349	1	-0.0348	-3.480
europe_in_aus_autumn_winter	-0.5279	1	-0.0344	-3.444
has_australia	-0.5271	1	-0.0344	-3.440
quote_late_nov_early_dec	-0.4217	1	-0.0287	-2.872
is_very_precise_destination	0.2848	1	0.0259	2.590
has_usa	-0.3697	1	-0.0257	-2.571
snow_sport_in_canada	0.2394	1	0.0214	2.137
cruise_boost	-0.2711	1	-0.0196	-1.963
platformqw	0.2182	1	0.0193	1.931
has_indonesia	-0.2634	1	-0.0191	-1.913
uk_in_aus_winter	-0.2206	1	-0.0163	-1.630
age_and_relationshipCouple (30–39)	0.1850	1	0.0162	1.615

Variable	Beta	ΔX	Δp	Δp (pp)
has_india	-0.2081	1	-0.0155	-1.546
has_nz	-0.2048	1	-0.0152	-1.523
snow_sport_with_children	0.1704	1	0.0148	1.479
canada_in_aus_winter	-0.1958	1	-0.0146	-1.462
is_holidays	-0.1927	1	-0.0144	-1.441
specified_item_boost	-0.1917	1	-0.0143	-1.434
age_and_relationshipCouple (70+)	-0.1879	1	-0.0141	-1.407
rental_vehicle_boost	-0.1793	1	-0.0135	-1.348
is_south_america	-0.1737	1	-0.0131	-1.309
group_size	-0.1680	1	-0.0127	-1.269
age_and_relationshipOther	-0.1630	1	-0.0123	-1.233
has_malaysia	0.1433	1	0.0123	1.230
spain_in_aus_winter	-0.1395	1	-0.0107	-1.066
age_and_relationshipCouple (18–29)	0.1249	1	0.0106	1.064
has_spain	-0.1382	1	-0.0106	-1.056
italy_in_aus_winter	-0.1300	1	-0.0100	-0.997
is_weekend	0.1058	1	0.0089	0.894
age_and_relationshipCouple (60–69)	0.1036	1	0.0087	0.875
is_africa	-0.1094	1	-0.0085	-0.846
extreme_exchange_risk_30d	0.0955	1	0.0080	0.804
tds_distance_long	-0.0736	1	-0.0058	-0.578
quote_hour_19_22	0.0596	1	0.0049	0.494
age_and_relationshipCouple (40–49)	0.0583	1	0.0048	0.483
senior_ratio	-0.2291	0.1667	-0.0030	-0.304
price_per_person_per_day	-0.0359	1	-0.0029	-0.286
has_italy	-0.0356	1	-0.0028	-0.284
is_antarctica	0.0343	1	0.0028	0.281
num_boosts1	0.0340	1	0.0028	0.279
booking_advance_ratio	-0.0312	1	-0.0025	-0.249
age_and_relationshipFamily (Children >18)	-0.0234	1	-0.0019	-0.188
tds_distance_short	0.0106	1	0.0009	0.086
num_days	-0.0086	1	-0.0007	-0.069
has_france	0.0066	1	0.0005	0.054
tds_distance_medium	0.0058	1	0.0005	0.047
has_japan	-0.0024	1	-0.0002	-0.019

G Boosts and Behavioral Signal from EDA

Boosts are optional add-on covers that travelers can attach to the base policy. They act as a revealed-intent signal that people who actively customize cover are usually taking the trip seriously and are closer to purchase. Two consistent patterns appear in the dataset.

(a) More boosts → higher intent → higher conversion. Customers who select more than one boost tend to convert at a higher rate, consistent with the strong positive coefficient of **number of boost** in the model results. travelers adding cover are not casually browsing, but they are planning around specific risks. The highest conversion rates (CR) are observed for Existing Medical Condition Cover: 38.2% and the lowest is Extra Cancellation: 2.6% (See Table 3). This shows that health- and activity-aligned boosts capture confident travelers, while “cancellation”-type boosts signal uncertainty and correlate with non-conversion.

(b) Boost choice depends on traveler profile, destination, and season. Younger travelers (19–34) prefer activity-based boosts such as Snow Sports, Adventure Activities, Gadget Cover, and Motorcycle while older travelers (59–79) favor Existing Medical and Cruise Cover. Small groups (<4 travelers) and shorter trips add more boosts and convert more often, suggesting urgency and readiness. Japan, Canada, USA → Snow Sports, Bali / Indonesia → Motorcycle, Gadget, Australia, New Zealand, Oceania → Cruise Cover. Australian summer and autumn align with the northern winter, producing strong Snow Sports boost demand; spring corresponds to Motorcycle boosts. Combined effects reinforce conversion: Motorcycle in Bali (42.2%), Snow Sports in Japan (32.8%), and in Canada (28.2%).

Table 3: Conversion Rates by Cover Type

Cover Type	Conversion Rate (%)
Existing Medical Condition Cover	38.2
Motorcycle Cover	36.6
Gadget Cover	33.2
Adventure Activities	31.1
Snow Sports	30.4
Lower Conversion Boosts:	
Rental Vehicle	18.1
Specified Items	16.2
Cruise	12.1
Extra Cancellation	2.6

H Methodology of Average Marginal Effect (AME) and Return on Investment (ROI)

H.1 Average Marginal Effect (AME)

The AME quantifies how a one–unit change in a predictor x_j affects the *average predicted probability of conversion*, while holding all other variables constant.

For a logistic regression model, the predicted probability for observation i is given by:

$$p_i = \frac{1}{1 + e^{-z_i}}, \qquad \text{where } z_i = X_i\beta. \tag{12}$$

Marginal effect via partial derivative Let $z_i = X_i\beta$ and $p_i = \frac{1}{1+e^{-z_i}}$. Then

$$\frac{\partial p_i}{\partial z_i} = \frac{\partial}{\partial z_i}\left(\frac{1}{1 + e^{-z_i}}\right) = \frac{e^{-z_i}}{(1 + e^{-z_i})^2} = \left(\frac{1}{1 + e^{-z_i}}\right)\left(1 - \frac{1}{1 + e^{-z_i}}\right) \tag{13}$$

$$= p_i(1 - p_i). \tag{14}$$

By the chain rule,

$$\frac{\partial p_i}{\partial x_{ij}} = \frac{\partial p_i}{\partial z_i} \cdot \frac{\partial z_i}{\partial x_{ij}} = \left[p_i(1 - p_i)\right] \cdot \beta_j \quad \Longrightarrow \quad ME_{ij} = \frac{\partial p_i}{\partial x_{ij}} = p_i(1 - p_i)\,\beta_j. \tag{15}$$

Average marginal effect (AME)

$$AME_j = \frac{1}{N_j}\sum_{i=1}^{N_j} ME_{ij} = \frac{1}{N_j}\sum_{i=1}^{N_j} p_i(1 - p_i)\,\beta_j, \tag{16}$$

where N_j denotes the sample size (number of observations) for predictor j used in the estimation of its marginal effect.

Binary vs. continuous predictors (exact change for dummies) For a binary predictor $x_{ij} \in \{0, 1\}$, one may use the exact discrete change

$$\Delta p_{ij} = p\Big(X_i,\, x_{ij} = 1\Big) - p\Big(X_i,\, x_{ij} = 0\Big), \tag{17}$$

and then average Δp_{ij} over i to obtain an AME on the $0 \to 1$ change scale; the derivative formula above is a first-order approximation to this discrete effect.

Coverage Coverage is defined as the proportion of a predictor's sample size relative to the total number of observations (69,966), expressed as a percentage:

$$\text{Coverage} = \frac{\text{Sample size}}{69{,}966} \times 100\%. \tag{18}$$

H.2 Return on Investment (ROI)

To evaluate financial efficiency by segment, define

$$\Delta p_j = AME_j, \qquad ROI_j = \frac{N_j \times \Delta p_j \times R_j - C_j}{C_j}. \tag{19}$$

where

N_j: Number of customers influenced by predictor j, i.e. Coverage$_j$ × Total customers (= 69,966).

Δp_j: Average marginal effect for predictor j (in decimal, e.g. 0.021).

R_j: Average premium revenue per policy after deducting sales expenses and expected claims.

C_j: Marketing execution cost (e.g., ad spend, channel rebates, or tech investment).

Interpretation:

- $ROI_j > 0$: Profitable action; larger ROI_j implies higher marketing efficiency.
- $ROI_j < 0$: Inefficient investment; consider optimization or redirection.

Ranking predictors by ROI_j yields a priority map that connects AME uplift with economic impact.

H.3 Matrix of Each Quadrant: AME vs Customer Coverage)

See in the next page.

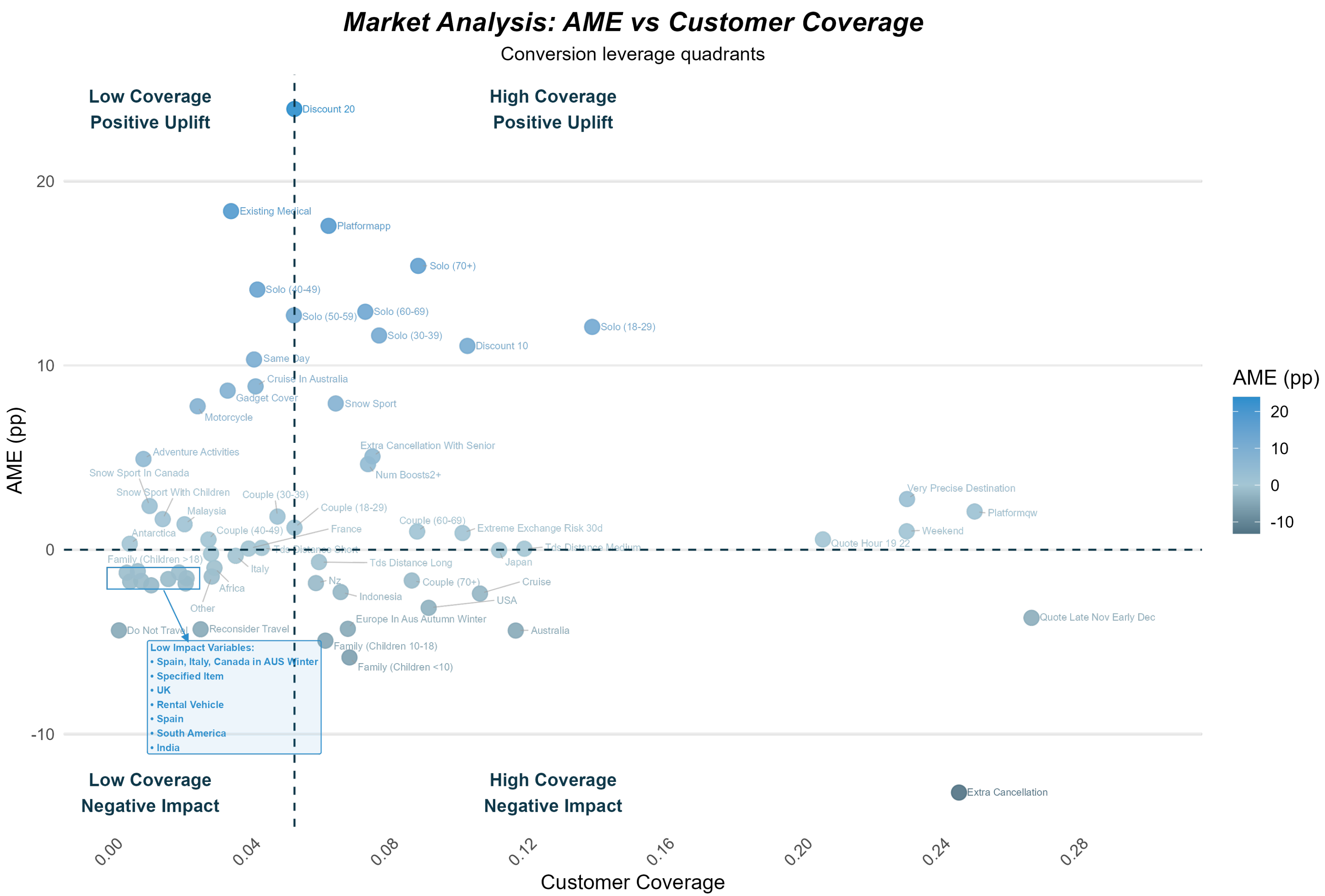

Figure 4: AME vs Customer Coverage Matrix

Note: The line in the figure represents the median sample size. Low and high coverage are defined as sample sizes below and above this median, respectively. All predictors in this plot are binary, while numerical predictors are listed separately in the following tables. AME (pp) denotes the Average Marginal Effect expressed in percentage points.

H.4 Table of Each Quadrant: AME vs Customer Coverage

H.4.1 Interpretation and usage:

- **Numerical predictors:** AME is the mean change in conversion probability for a +1 unit change (or +1/6 increment for ratio variables such as child_ratio and senior_ratio).
- **Binary predictors:** AME is the mean change in predicted conversion when the variable changes from $0 \to 1$.
- **Sign and magnitude:** Positive AME $\Rightarrow$ higher conversion likelihood; negative AME $\Rightarrow$ lower likelihood; magnitude reflects effect strength (scale-free).

When combined with customer coverage, the AME–Coverage matrix classifies marketing segments into four quadrants:

Quadrant	Interpretation
I. High AME / High Coverage	Stable strengths
II. High AME / Low Coverage	Growth opportunities
III. Low AME / Low Coverage	Low priority
IV. Low AME / High Coverage	Optimization targets

H.4.2 Numerical Predictors

For numerical predictors, the AME measures how conversion probability changes when the variable increases by a defined increment.

Predictor	Type	Increment in Predictor	AME (pp) Change
child_ratio	continuous	+1/6	+4.63
senior_ratio	continuous	+1/6	−0.35
group_size	discrete	+1	−1.49
price_per_day	continuous	+1	−0.33
num_days	discrete	+1	−0.08

Example: For child_ratio, an increase of roughly one additional child in the quote (+1/6) raises the expected conversion rate by 4.63 pp.

H.4.3 Binary Predictors

For binary predictors, AME is the change in conversion probability when the variable switches from 0 to 1. Coverage is each predictor's sample size divided by 69,966.

Example. For platform_app (Coverage = 6.20%, AME = +17.57 pp), when a customer uses the app to obtain a quote, the expected conversion rate increases by about 17.57 percentage points relative to other platforms.

Quadrant I — High Coverage / Positive AME (Upper Right)

Predictors	Coverages	AME (pp)
platformapp	6.20%	+17.57
Solo (70+)	8.80%	+15.4
Solo (60–69)	7.20%	+12.91
discount_15	45.80%	+12.25
Solo (18–29)	13.80%	+12.09
Solo (30–39)	7.60%	+11.62
discount_10	10.20%	+11.06
snow_sport_boost	6.40%	+7.93
extra_cancellation_with_senior	7.40%	+5.06
num_boosts2+	7.30%	+4.64
is_very_precise_destination	22.90%	+2.75
platformqw	24.90%	+2.07
Couple (18–29)	5.20%	+1.2
is_weekend	22.90%	+1
Couple (60–69)	8.70%	+0.99
extreme_exchange_risk_30d	10.10%	+0.91
quote_hour_19_22	20.50%	+0.56
num_boosts1	39.20%	+0.32
tds_distance_medium	11.80%	+0.05

Quadrant II — Low Coverage / Positive AME (Upper Left)

Predictor	Coverages	AME (pp)
discount_20	5.20%	+23.91
existing_medical_boost	3.30%	+18.37
Solo (40–49)	4.10%	+14.12
Solo (50–59)	5.20%	+12.71
Same_day	4.00%	+10.32
cruise_in_australia	4.10%	+8.86
gadget_cover_boost	3.20%	+8.63
motorcycle_boost	2.40%	+7.78
adventure_activities_boost	0.80%	+4.92
snow_sport_in_canada	1.00%	+2.37
Couple (30–39)	4.70%	+1.8
snow_sport_with_children	1.40%	+1.66
has_malaysia	2.00%	+1.38
Couple (40–49)	2.70%	+0.55
is_antarctica	0.40%	+0.32
tds_distance_short	4.20%	+0.1
has_france	3.90%	+0.06

Quadrant III — Low Coverage / Negative AME (Lower Left)

Predictor	Coverages	AME (pp)
warning_do_not_travel	0.10%	-4.37
warning_reconsider_travel	2.50%	-4.31
uk_in_aus_winter	1.00%	-1.93
has_india	2.00%	-1.83
canada_in_aus_winter	0.40%	-1.73
specified_item_boost	0.70%	-1.69
rental_vehicle_boost	1.50%	-1.59
is_south_america	2.10%	-1.54
relationship_Other	2.80%	-1.45
spain_in_aus_winter	0.30%	-1.25
has_spain	1.80%	-1.24
italy_in_aus_winter	0.60%	-1.17
is_africa	2.90%	-0.99
has_italy	3.50%	-0.33
Family (Children > 18)	2.80%	-0.22

Quadrant IV — High Coverage / Negative AME (Lower Right)

Predictor	Coverages	AME (pp)
extra_cancellation_boost	24.50%	-13.17
Family (Children < 10)	6.80%	-5.85
Family (Children 10–18)	6.10%	-4.93
has_australia	11.60%	-4.38
europe_in_aus_autumn_winter	6.70%	-4.29
quote_late_nov_early_dec	26.60%	-3.7
has_usa	9.10%	-3.15
cruise_boost	10.60%	-2.38
has_indonesia	6.50%	-2.3
has_nz	5.80%	-1.81
is_holidays	55.30%	-1.8
Couple (70+)	8.60%	-1.67
tds_distance_long	5.90%	-0.67
has_japan	11.10%	-0.02